

QA Documentation: Online Course Helpers Website

Date: January 16, 2026
Status: Code Review & Bug Identification

Summary

This document provides a comprehensive QA review of the Online Course Helpers website, identifying all bugs, issues, and areas requiring fixes across PHP, JavaScript, and CSS files.

CRITICAL ISSUES

1. index.php

Issue 1.1: Empty PHP Tag

Severity: △ Low | **Type:** Code Quality
Location: Lines 1-3

```
<?php
// index.php
?>
```

Problem: The opening PHP tag contains only a comment with no functional code.
Impact: Unnecessary overhead; confusing for developers.
Fix: Remove empty PHP tags since no server-side processing occurs on this page.

Issue 1.2: Dead Navigation Links

Severity: ⬢ High | **Type:** Functionality
Location: Lines 44-60

```
<!-- Navigation Items -->
<div class="collapse navbar-collapse" id="navbarNav">
  <ul class="navbar-nav mx-auto gap-3 align-items-center">
    <li class="nav-item">
      <a class="nav-link active" href="#">Home</a>      <!-- ⬢ DEAD LINK -->
    </li>
    <li class="nav-item">
      <a class="nav-link" href="#">Services</a>      <!-- ⬢ DEAD LINK -->
    </li>
    <li class="nav-item">
      <a class="nav-link" href="#">Platforms</a>      <!-- ⬢ DEAD LINK -->
    </li>
    <li class="nav-item">
      <a class="nav-link" href="#">Courses</a>      <!-- ⬢ DEAD LINK -->
    </li>
    <li class="nav-item">
      <a class="nav-link" href="#">Education Level</a>      <!-- ⬢ DEAD LINK -->
    </li>
    <li class="nav-item">
      <a class="nav-link" href="#">Blog</a>      <!-- ⬢ DEAD LINK -->
    </li>
```

Problem: All navigation links point to # (nowhere). Navigation will not work.
Impact: Users cannot navigate to different sections of the website.
Fix: Replace href="#" with proper anchors like href="#services", href="#platforms", etc. and ensure corresponding section IDs exist in the HTML.

MAJOR ISSUES

2. hero_section.php

Issue 2.1: No Form Validation & Backend Processing

Severity: 🚨 High | **Type:** Security & Functionality
Location: Lines 49-73

```
<form id="contactForm">
  <div class="mb-3">
    <label for="name" class="form-label">Name</label>
    <input type="text" class="form-control" id="name" placeholder="Name" required />
    <!-- 🚨 No pattern validation, no sanitization -->
  </div>
  <div class="mb-3">
    <label for="email" class="form-label">Email</label>
    <input type="email" class="form-control" id="email" placeholder="Email" required />
    <!-- 🚨 No server-side validation, vulnerable to injection attacks -->
  </div>
  <div class="mb-3">
    <label for="phone" class="form-label">Phone No</label>
    <input type="tel" class="form-control" id="phone" placeholder="Phone No" required />
    <!-- 🚨 No international format support, no backend processing -->
  </div>
  <div class="mb-4">
    <label for="service" class="form-label">Services</label>
    <select class="form-select" id="service" required>
      <option value="" disabled selected>Select Service</option>
      <option value="1">Online Class</option>
      <option value="2">Online Course</option>
      <option value="3">Assignments</option>
      <option value="4">Dissertation</option>
    </select>
  </div>
  <button type="submit" class="btn btn-submit">
    Send
    <!-- 🚨 No CSRF token for security -->
  </button>
</form>
```

Problems:

- No backend PHP script to process the form submission
- Missing CSRF token for security
- No input sanitization or validation
- No email verification
- Potential SQL injection vulnerability
- No error handling

Impact: Form data is lost; users have no feedback; security vulnerabilities exposed.
Fix:

- Create a backend handler (e.g., process_form.php)
- Add CSRF token to form
- Implement server-side validation and sanitization
- Add error handling and user feedback

Issue 2.2: Missing Alt Text on Images

Severity: ⚠ Medium | **Type:** Accessibility & SEO
Location: Line 37

```

<!-- ✓ This one has proper alt text -->
```

Problem: While this specific image has alt text, it's not descriptive enough for SEO purposes.
Impact: Poor SEO; accessibility issues for screen readers.

3. benefits_section.php

Issue 3.1: Duplicate Course Cards (Major Content Issue)

Severity: 🚨 High | **Type:** Content & User Experience
Location: Lines 44-80+

```
<!-- Card 1: Finance Course Help -->
<div class="course-card-container">
  <div class="course-card">
    <h3>Finance Course Help</h3>
    <!-- Content... -->
  </div>
</div>

<!-- Card 2: Business Law Help -->
<div class="course-card-container">
  <div class="course-card">
    <h3>Business Law Help</h3>
    <!-- Content... -->
  </div>
</div>

<!-- Card 3: Statistics Help -->
<div class="course-card-container">
  <div class="course-card">
    <h3>Statistics Help</h3>
    <!-- Content... -->
  </div>
</div>

<!-- 🚨 DUPLICATE: Statistics Help (Exact Copy) -->
<div class="course-card-container">
  <div class="course-card">
    <h3>Statistics Help</h3>
    <p>We help with probability, data analysis, testing hypotheses and so on...</p>
    <!-- IDENTICAL TO CARD 3 -->
  </div>
</div>

<!-- 🚨 DUPLICATE: Statistics Help (Exact Copy #2) -->
<div class="course-card-container">
  <div class="course-card">
    <h3>Statistics Help</h3>
    <p>We help with probability, data analysis, testing hypotheses and so on...</p>
    <!-- IDENTICAL TO CARD 3 -->
  </div>
</div>

<!-- 🚨 DUPLICATE: Statistics Help (Exact Copy #3) -->
<div class="course-card-container">
  <div class="course-card">
    <div class="card-content">
      <h3>Statistics Help</h3>
      <p>We help with probability, data analysis, testing hypotheses and so on...</p>
    </div>
  </div>
</div>
```

Problem:

- 3 identical "Statistics Help" cards are repeated
 - Wastes horizontal scroll space
 - Poor user experience
 - Appears unprofessional
- Impact:** Users see redundant content; confusing navigation; reduced trust.
Fix: Replace duplicate cards with unique course offerings:

- Biology Course Help
- Chemistry Course Help
- Economics Course Help
- Literature Course Help
- Computer Science Course Help
- Psychology Course Help

4. cheap_price_section.php

Issue 4.1: Empty Section

Severity: 🚨 High | **Type:** Incomplete Feature
Location: Entire file

```
<section class="cheap-price-section"></section>
```

Problem: Section is completely empty; no content is rendered.
Impact: Users expect pricing information; section title suggests pricing content should exist.
Fix: Implement pricing table or pricing tier cards with:

- Basic, Standard, Premium plan tiers
- Feature comparisons
- Price points
- Call-to-action buttons

5. moodle_section.php

Issue 5.1: Empty Alt Attributes

Severity: ⚠ Medium | **Type:** Accessibility & SEO
Location: Lines 18, 30, 40, 50, 60, 70, 80+ (Multiple instances)

```

<!-- 🚨 Empty alt attribute -->


<!-- 🚨 Empty alt attribute -->


<!-- 🚨 Empty alt attribute -->
```

Problem: Empty `alt=""` attributes provide no alternative text for screen readers or when images fail to load.
Impact: Accessibility issues; poor SEO; fail WCAG standards.
Fix: Add descriptive alt text:

```



```

Issue 5.2: Duplicate ID Attribute

Severity: ⚠ Medium | **Type:** HTML Standards Violation
Location: Multiple images (Lines 18, 30, 40, 50, etc.)

```

<!-- First instance -->


<!-- 🚨 DUPLICATE ID: Should be unique! -->


<!-- 🚨 DUPLICATE ID: Should be unique! -->
```

Problem: HTML IDs must be unique within a page. Having multiple elements with the same ID violates HTML standards and breaks CSS/JavaScript selectors.
Impact: CSS and JavaScript targeting may only affect the first element; unpredictable behavior; hard to debug.
Fix: Use classes instead or create unique IDs:

```
<!-- Option 1: Use class instead -->



<!-- Option 2: Create unique IDs -->


```

🔗 CSS ISSUES

6. styles.css

Issue 6.1: Invalid Gradient Color Stop Order

Severity: ⚠ Medium | **Type:** CSS Error

Location: Lines 25-32

```
.hero-section::before {
  position: absolute;
  top: 0;
  bottom: 0;
  left: 0;
  right: 0;
  content: "";
  background: linear-gradient(135deg,
    #003d4d 0%,
    #003042 60%,          /* ❌ 60% */
    rgb(0, 39, 54) 20%,   /* ❌ 20% (WRONG ORDER! Should be after 60%) */
    #f3ef00 100%);
  z-index: 1;
  opacity: 0.95;
}
```

Problem: Gradient color stops are out of order (60% before 20%). This is illogical and may cause unexpected gradient rendering.

Impact: Incorrect visual rendering; gradient may not display as intended.

Fix: Arrange color stops in ascending order:

```
background: linear-gradient(135deg,
  #003d4d 0%,
  #003042 20%,          /* ✓ Corrected order */
  rgb(0, 39, 54) 60%,   /* ✓ Corrected order */
  #f3ef00 100%);
```

Issue 6.2: SCSS Syntax in Regular CSS File

Severity: ⚠ Medium | **Type:** CSS Compilation Error

Location: Lines 283-294

```
.slider-images {
  display: flex;
  gap: 10px;
  overflow-x: auto;
  white-space: nowrap;

  /* Hide scrollbar for Chrome, Safari, Edge, and Opera */
  &::-webkit-scrollbar {          /* ❌ SCSS Syntax! Not valid CSS */
    display: none;
  }

  /* Hide scrollbar for Firefox */
  scrollbar-width: none;

  /* Hide scrollbar for IE and older Edge */
  -ms-overflow-style: none;
}
```

Problem: The `&::-webkit-scrollbar` selector uses SCSS nesting syntax (`&`), which is invalid in standard CSS.

Impact: This CSS rule will not work; scrollbar styling will fail.

Fix: Convert to standard CSS:

```
.slider-images {
  display: flex;
  gap: 10px;
  overflow-x: auto;
  white-space: nowrap;
  scrollbar-width: none;
  -ms-overflow-style: none;
}

/* Hide scrollbar for Chrome, Safari, Edge, and Opera */
.slider-images::-webkit-scrollbar {
  display: none;
}
```

Issue 6.3: Duplicate `.highlight` Class Definition

Severity: △ Low | **Type:** CSS Maintainability

Location: Lines 244 and 318

```
/* First definition (Line 244) */
.highlight {
  background-color: #ffd000;
  padding: 2px 6px;
}

/* ... other rules ... */

/* Second definition (Line 318) - DUPLICATE */
.highlight {
  background-color: #FFCC00;
  color: #000;
  padding: 2px 10px;
}
```

Problem: Two conflicting definitions of the same class. The second one will override the first.

Impact: Confusion; potential CSS conflicts; hard to maintain.

Fix: Create separate classes or merge the definitions:

```
.highlight {
  background-color: #FFCC00;
  color: #000;
  padding: 2px 10px;
}

/* Remove the first definition or rename it */
```

Issue 6.4: Missing Shadow Effect on Hover

Severity: △ Low | **Type:** UX Enhancement

Location: Lines 512-516

```
.service-box {
  background-color: #00334d;
  color: #ffffff;
  padding: 40px 30px;
  border-radius: 8px;
  height: 100%;
  transition: transform 0.3s ease;
  /* ❌ No box-shadow defined */
}

.service-box:hover {
  transform: translateY(-5px);
  /* ❌ No enhanced shadow on hover */
}
```

Problem: Hover effect only applies `translateY` but no shadow enhancement. This looks incomplete compared to modern UI patterns.

Impact: Less polished visual feedback; inconsistent with contemporary design standards.

Fix: Add shadow effects:

```
.service-box {
  /* ... existing styles ... */
  box-shadow: 0 2px 8px rgba(0, 0, 0, 0.1);
  transition: transform 0.3s ease, box-shadow 0.3s ease;
}

.service-box:hover {
  transform: translateY(-5px);
  box-shadow: 0 8px 16px rgba(0, 0, 0, 0.2);
}
```

Issue 6.5: Repeated CSS Rule Definition

Severity: △ Low | **Type:** Code Redundancy

Location: Lines 400-415 and 506-520

```
/* First definition */
.horizontal-scroll-wrapper::-webkit-scrollbar {
  height: 8px;
}

.horizontal-scroll-wrapper {
  scrollbar-color: #00334d #FFE066;
  scrollbar-width: thin;
}

/* Later in file - essentially repeats the scrollbar styling */
```

Problem: Similar scrollbar styling rules appear in multiple places.

Impact: Code bloat; harder to maintain.

Fix: Consolidate repeated rules into a single location.

📄 JAVASCRIPT ISSUES

7. app.js

Issue 7.1: Inefficient Scroll Event Listener

Severity: ⚠ Medium | **Type:** Performance

Location: Lines 53-63

```
$(window).scroll(function () {
  if ($(this).width() > 992) {
    $(".navbar").css("background-color", "#111 !important");
    /* 🐛 This runs on EVERY pixel scrolled - very inefficient! */
  }
  if ($(this).scrollTop() > 10) {
    $(".navbar").addClass("scrolled");
  } else {
    $(".navbar").removeClass("scrolled");
  }
});
```

Problems:

1. Scroll event fires hundreds of times per second (on every pixel)
2. DOM manipulation (`.css()`) happens on every scroll event
3. Inline `!important` flag is used (CSS anti-pattern)
4. No throttling or debouncing
5. Width check runs unnecessarily on every scroll

Impact:

- Performance degradation on lower-end devices
- Excessive CPU usage
- Battery drain on mobile devices
- Janky scrolling experience

Fix: Implement throttling and optimize the code:

```
let lastScrollTime = 0;
const throttleDelay = 100; // milliseconds

$(window).scroll(function () {
  const currentTime = Date.now();

  // Throttle scroll event
  if (currentTime - lastScrollTime < throttleDelay) return;
  lastScrollTime = currentTime;

  // Move width check outside scroll handler - only check once
  if ($(this).scrollTop() > 10) {
    $(".navbar").addClass("scrolled");
  } else {
    $(".navbar").removeClass("scrolled");
  }
});

// Check width only on window resize, not on scroll
$(window).resize(function () {
  if ($(this).width() > 992) {
    $(".navbar").removeClass("scrolled-mobile");
  } else {
    $(".navbar").addClass("scrolled-mobile");
  }
});
```


Issue 7.2: Ambiguous Variable Names

Severity: △ Low | Type: Code Quality
Location: Lines 30-40

```
$horizontalScrollCards.on("click", () => {  
  let direction = 1;  
  const value = 300; /* ❓ What is 'value'? Pixels? Milliseconds? Unclear */  
  const currentScroll = $horizontalScrollWrapper.scrollLeft();  
  
  const maxScroll = $horizontalScrollWrapper[0].scrollWidth - $horizontalScrollWrapper.outerWidth();  
  
  if (currentScroll >= maxScroll - 5)  
    direction = -1;  
  else if (currentScroll <= 0) {  
    direction = 1;  
  }  
})
```

Problem: Variable names are vague; poor documentation.
Impact: Hard to maintain; confusing for other developers.
Fix: Use descriptive names:

```
const SCROLL_DISTANCE = 300; // pixels  
const ANIMATION_DURATION = 300; // milliseconds  
const SCROLL_END_THRESHOLD = 5; // pixels
```

📄 SUMMARY TABLE

Issue ID	File	Severity	Type	Status
1.1	index.php	Low	Code Quality	Not Started
1.2	index.php	High	Functionality	Not Started
2.1	hero_section.php	High	Security	Not Started
2.2	hero_section.php	Medium	Accessibility	Not Started
3.1	benefits_section.php	High	Content	Not Started
4.1	cheap_price_section.php	High	Incomplete	Not Started
5.1	moodle_section.php	Medium	Accessibility	Not Started
5.2	moodle_section.php	Medium	HTML Violation	Not Started
6.1	styles.css	Medium	CSS Error	Not Started
6.2	styles.css	Medium	CSS Syntax	Not Started
6.3	styles.css	Low	Maintenance	Not Started
6.4	styles.css	Low	UX Enhancement	Not Started

Issue ID	File	Severity	Type	Status
6.5	styles.css	Low	Code Redundancy	Not Started
7.1	app.js	Medium	Performance	Not Started
7.2	app.js	Low	Code Quality	Not Started

📌 Recommended Fix Priority

📌 Priority 1: CRITICAL (Fix Immediately)

- 1. Issue 1.2: Dead navigation links
- 2. Issue 2.1: Form validation & backend processing
- 3. Issue 3.1: Duplicate course cards
- 4. Issue 4.1: Empty pricing section

📌 Priority 2: IMPORTANT (Fix Soon)

- 1. Issue 6.1: Invalid gradient color stops
- 2. Issue 6.2: SCSS syntax in CSS
- 3. Issue 7.1: Inefficient scroll event
- 4. Issue 5.1: Empty alt attributes

📌 Priority 3: NICE-TO-HAVE (Fix When Time Permits)

- 1. Issue 6.3: Duplicate .highlight class
- 2. Issue 6.4: Missing shadow effects
- 3. Issue 7.2: Ambiguous variable names

📌 Notes

- All issues documented with code snippets for easy reference
 - Fixes suggested for each issue
 - Testing recommended after each fix
 - Consider implementing linting tools (ESLint, StyleLint) to prevent future issues
-

End of QA Documentation